

FYP: Comparative Analysis of Data Center Cooling Strategies

Simulation Results & Analysis Report

Aleeza Hussain

April 2026

This report presents the full simulation results from three high-fidelity physics-based cooling models: Air-Side Economization, Evaporative Cooling, and Chilled Water Systems. All simulations run over 8,760 hours (one full year) using CloudSim Plus 8.x co-simulation integrated with custom Spring Boot / Python physics engines.

Department of Computer Science | Final Year Project

1. Executive Summary

This report presents the complete simulation results for three data center cooling strategies evaluated under identical AI-era workload conditions over a full 8,760-hour annual cycle. The simulation framework integrates CloudSim Plus 8.x Discrete Event Simulation with custom-built physics engines implemented in Java 17 / Spring Boot 3.x and Python FastAPI, as described in the accompanying Methodology document.

Key Findings at a Glance

Metric	Air-Side Econ.	Evaporative	Chilled Water	Best
Total Energy (kWh/yr)	478,090	141,539	288,475	Evaporative
Average PUE	1.285	1.002	1.163	Evaporative
Average CUE (kgCO2/kWhIT)	0.591	0.461	0.549	Evaporative
Annual OpEx (USD)	\$97,033	\$20,523	\$42,553	Evaporative
Carbon Emissions (kg/yr)	219,921	65,108	158,196	Evaporative
Water Usage (L/yr)	0	0	515,331	Air/Evap
CAPEX (USD)	\$45,000	Low	\$630,000	Evaporative
Thermal Compliance	Marginal	FAIL	PASS	Chilled Water
Payback Period (yrs)	491.9	N/A	50.0	---

Table 1: Annual Performance Summary - All Three Cooling Techniques

Overall Recommendation: The ML-based recommender system recommends **Evaporative Cooling** for lowest energy and cost, but the system carries a critical thermal compliance failure - rack inlet temperatures reached 38.5 degrees C, exceeding the ASHRAE Class A1 limit of 27 degrees C. The **Chilled Water System** is the only thermally compliant solution for high-density AI workloads and scores highest (0.930) in the ML composite ranking.

2. Simulation Architecture and Co-Simulation Design

The simulation platform is built on a lock-step co-simulation model that tightly couples a Discrete Event Simulation (DES) layer with a physics-based thermal engine. The simulation clock advances in hourly increments ($\Delta t = 3,600\text{ s}$). At each tick, the Thermal Layer pulls P_{IT} from the Computational Layer, calculates the necessary mass flow rate of air or water, and returns the parasitic cooling power ($P_{cooling}$) back to the global power tracker.

2.1 Co-Simulation Layer Responsibilities

Layer	Technology	Responsibility
Computational	CloudSim Plus 8.x	VM provisioning, task scheduling, hourly P_{IT} output
Thermal	Spring Boot 3.x / Java 17	Physics engine: airflow, heat balance, COP calculations
Synchronisation	Lock-step ($\Delta t=3600s$)	Pulls P_{IT} each tick; returns $P_{cooling}$ to global tracker
Economic	ProjectionEngine.java	5-year NPV, carbon tax escalation, TOU tariffs
ML Recommender	Random Forest (scikit-learn)	Ranks techniques by composite score

Table 2: Co-Simulation Layer Responsibilities

2.2 Workload Model

All three simulations use an identical AI-aware composite workload model: $U_{total}(t) = [U_{diurnal}(t) \times M_{AI}] + \sigma(t)$, where $U_{diurnal}$ is a sinusoidal function peaking at 14:00, $M_{AI} = 1.3$ (Mixed AI workload), and $\sigma(t)$ is Gaussian noise (+/-5%). Weekend load is reduced by 70%. UPS and PDU losses are modelled with quadratic efficiency curves: $\eta = -0.05u^2 + 0.1u + 0.94$.

2.3 Common Simulation Parameters

Category	Parameter	Value	Source
Thermal	ASHRAE Max Inlet (T_{max})	27.0 deg C	ASHRAE TC 9.9
Thermal	Thermal Mass (Rack)	15.0 kJ/K	Design Default
Psychrometric	Air Density (ρ)	1.2 kg/m3	Ideal Gas Law
Psychrometric	Latent Heat (L_v)	2,260 kJ/kg	Water Vaporisation
Power	Chiller Reference COP	3.0	Conservative Baseline
Economic	Base Electricity Rate	\$0.12/kWh	Regional Default
Carbon	Carbon Tax (2030)	\$254/ton CO2	IPCC Pathway
Workload	AI Training Multiplier	1.8x	Sustained High Load
Control	Mode Switch Hysteresis	2.0 deg C	Stability Buffer
Simulation	Total Hours	8,760	Full Annual Cycle

Table 3: Common Simulation Input Parameters

3. Air-Side Economizer: Simulation Results

3.1 System Description

The Air-Side Economizer leverages ambient outdoor air to offset mechanical cooling. A decision-tree state machine selects between three operating modes based on outdoor dry-bulb temperature (T_{out}) and relative humidity (RH_{out}). Required volumetric airflow is: $V_{req} \text{ (CFM)} = (P_{IT_kW} \times 3160) / (\rho \times C_p \times \Delta T)$. A 15% filter penalty is applied to fan power in economiser modes (MERV filtration resistance). The three operating modes are: FULL_ECON ($T_{out} \leq 24$ deg C and $RH \leq 60\%$, compressor off), PARTIAL_TRIM ($T_{out} \leq 24$ deg C but $RH > 60\%$, 20% outdoor air), and MECHANICAL_ONLY ($T_{out} > 24$ deg C, full compressor operation).

3.2 Annual Energy Results (8,760-Hour Simulation)

Metric	Value	Unit
Total IT Energy	372,125	kWh
Total Cooling Energy	105,965	kWh
Total Facility Energy	478,090	kWh
Average PUE	1.285	-
Average CUE	0.591	kgCO2/kWhIT
Total Carbon Emissions	219,921	kg CO2
Carbon Savings vs. Baseline	88,198	kg CO2
Energy Savings vs. Baseline	28.6%	-
Water Usage	0	Litres

Table 4: Air-Side Economizer - Annual Energy & Emissions Summary

3.3 Economic Results

Metric	Value	Unit
CAPEX	\$45,000	USD
Annual Electricity Cost	\$69,323	USD
Annual Carbon Tax Cost	\$27,710	USD
Total Annual OpEx	\$97,033	USD
Annual Savings vs. Baseline	\$91	USD
Simple Payback Period	491.9	Years
NPV of Savings (5-yr, 9.71% discount)	\$81,161	USD
Adjusted Payback (demo scenario \$25k CAPEX)	1.31	Years

Table 5: Air-Side Economizer - Economic Analysis

Note on Payback Period: The 491.9-year payback reflects the high carbon tax burden (\$27,710/yr) at the IPCC 2030 rate of \$254/ton CO₂. The system's carbon emissions (219,921 kg/yr) are high because the simulation site uses a carbon-intensive grid (0.55 kgCO₂/kWh). The adjusted payback of 1.31 years applies to the lower-CAPEX demo scenario (\$25,000 CAPEX) where carbon tax is not included.

3.4 5-Year Financial Projection

Year	Energy (kWh)	Elec. Cost	Carbon Tax	Total Cost	Cum. Savings
1	435,449	\$65,317	\$3,018	\$68,335	\$18,856
2	435,449	\$67,603	\$3,470	\$71,074	\$38,468
3	435,449	\$69,970	\$3,991	\$73,960	\$58,877
4	435,449	\$72,419	\$4,589	\$77,008	\$80,126
5	435,449	\$74,953	\$5,278	\$80,231	\$102,265
TOTAL	2,177,245	\$350,262	\$20,346	\$370,608	\$102,265

Table 6: Air-Side Economizer - 5-Year Cost Projection (15% Carbon Tax Escalation)

3.5 CloudSim Rack-Level Analysis

The CloudSim integration revealed critical airflow violations under AI workloads. Average rack utilisation was 78.2% (Mixed AI workload). Load imbalance factor CV = 0.30 indicates significant workload skew across racks.

Rack	Peak Load (kW)	Required CFM	Limit (CFM)	Status
Rack 0	5.5	803	400	VIOLATION
Rack 1	5.4	792	400	VIOLATION
Rack 2	5.5	809	400	VIOLATION
Rack 3	10.0	1,464	400	HOTSPOT + VIOLATION
Rack 4	10.0	1,456	400	HOTSPOT + VIOLATION
Average	7.05	-	-	5/5 racks violated

Table 7: Air-Side Economizer - Rack-Level Hotspot Analysis (Mixed AI Workload)

All five racks exceed the 400 CFM per-rack airflow limit. Racks 3 and 4 are classified as hotspots (peak load 33% above the 7.5 kW threshold). The system recommends upgrading to liquid cooling (direct-to-chip) for AI-density workloads.

3.6 Annual Mode Distribution

Mode	Approx. Hours	% of Year	Avg PUE	Description
FULL_ECON	~3,500	~40%	1.23	Compressor off, 100% outdoor air
PARTIAL_TRIM	~1,750	~20%	1.33	20% outdoor air, partial mechanical
MECHANICAL_ONLY	~3,510	~40%	1.45	Full compressor, 0% outdoor air
Annual Average	8,760	100%	1.285	-

Table 8: Air-Side Economizer - Annual Operating Mode Distribution

3.7 Phase Gate Assessment

Gate	Status	Notes
Thermal Compliance	MARGINAL	Airflow violations at all racks under AI load
Water Constraint	PASS	Zero water consumption
Carbon Liability	FAIL	219,921 kg CO2/yr; high grid carbon intensity
Economic Viability	FAIL	Payback 491.9 yrs at full CAPEX (\$45k)

Table 9: Air-Side Economizer - Phase Gate Compliance

4. Evaporative Cooling: Simulation Results

4.1 System Description

The evaporative cooling system leverages the latent heat of vaporisation ($L_v = 2,260 \text{ kJ/kg}$) to reduce supply air temperature. The simulation implements Direct Evaporative Cooling (DEC) and Indirect Evaporative Cooling (IEC) modes. Supply temperature is modelled as: $T_{\text{supply}} = T_{\text{db}} - \eta_{\text{actual}} \times (T_{\text{db}} - T_{\text{wb}})$, where η_{actual} is the saturation effectiveness (60-95%). A lumped capacitance model prevents thermal spikes: $C_{\text{total}} \times dT/dt = Q_{\text{IT}}(t) - Q_{\text{cooling}}(t)$, where $C_{\text{total}} = C_{\text{rack}} + C_{\text{enclosure}} = 45 \text{ kJ/K}$.

4.2 Annual Energy Results (8,760-Hour Simulation)

Metric	Value	Unit
Total IT Energy	141,173	kWh
Fan Energy (Auxiliary)	366	kWh
DX Compressor Energy	0	kWh
Total Facility Energy	141,539	kWh
Average PUE	1.002	-
Peak PUE	1.004	-
Average CUE	0.461	kgCO2/kWhIT
Total Carbon Emissions	65,108	kg CO2
Water Usage	0	Litres
Cooling Failure Hours	0	Hours

Table 10: Evaporative Cooling - Annual Energy & Emissions Summary

The evaporative system achieves a near-ideal PUE of 1.002, meaning only 0.2% overhead above the IT load - the best result of all three techniques. Zero DX compressor energy is consumed, and the system operates entirely on fan power.

4.3 Economic Results

Metric	Value	Unit
Annual Electricity Cost	\$20,523	USD
Water Cost	\$0	USD
Total Annual OpEx	\$20,523	USD
OpEx per kWhIT	\$0.1454	USD/kWh
OpEx per Server (Annual)	\$213.78	USD
Carbon Emissions	65,108	kg CO2
CO2 per kWhIT	0.461	kg/kWh

Table 11: Evaporative Cooling - Economic Analysis

The evaporative system has the lowest annual OpEx of all three techniques at \$20,523/yr - 79% lower than Air-Side Economizer and 52% lower than Chilled Water.

4.4 Sample Hourly Performance Profile

Hour	Amb. Temp (C)	IT Load (kW)	Fan Power (kW)	Total (kW)	PUE	Mode
0	25.0	11.760	0.0142	11.774	1.0012	DEC
1	25.0	11.760	0.0142	11.774	1.0012	DEC
5	25.0	11.798	0.0144	11.813	1.0012	DEC
12	25.0	12.805	0.0184	12.823	1.0014	DEC
14	25.0	13.089	0.0196	13.109	1.0015	DEC
18	25.0	13.032	0.0194	13.052	1.0015	DEC
23	25.0	11.808	0.0144	11.823	1.0012	DEC

Table 12: Evaporative Cooling - Sample Hourly Results

4.5 Cooling Assessment and Thermal Compliance

Check	Result	Detail
Heat Balance	FAIL	Cooling capacity 3.5% below IT heat load at peak
Inlet Temperature	FAIL	Max inlet temp: 38.5 deg C (ASHRAE limit: 27 deg C)
Humidity	PASS	Supply humidity within acceptable range
Energy Efficiency	PASS	PUE 1.002 - excellent
Overall Status	INSUFFICIENT_COOLING	Confidence: 95%

Table 13: Evaporative Cooling - Cooling Assessment Checks

Critical Finding: Rack inlet temperatures reached 38.5 deg C, exceeding the ASHRAE Class A1 limit of 27 deg C by 11.5 deg C. This triggers the thermal throttling model: $\text{Phi_adj} = \text{Phi_base} \times (1 - \text{gamma} \times \text{delta_T})$. At $\text{delta_T} = 11.5$ deg C and $\text{gamma} = 0.10$ (aggressive throttling above 29 deg C), computational throughput is reduced by up to 115% - effectively a full shutdown. This makes standalone evaporative cooling unsuitable for high-density AI workloads without supplemental cooling.

4.6 Engineering Recommendations

- Increase evaporative airflow by ~5% to close the 3.5% capacity gap.
- Reduce IT load density or improve server efficiency.
- Deploy as a pre-cooling stage upstream of a chilled water system (hybrid configuration).
- Monitor dewpoint limits carefully - direct evaporative cooling adds humidity to supply air.

4.7 Phase Gate Assessment

Gate	Status	Notes
Thermal Compliance	FAIL	Inlet temp 38.5 deg C; 11.5 deg C above ASHRAE limit
Water Constraint	PASS	Zero water consumption in IEC mode
Carbon Liability	PASS	Lowest emissions: 65,108 kg CO2/yr
Economic Viability	PASS	Lowest OpEx: \$20,523/yr

Table 14: Evaporative Cooling - Phase Gate Compliance

5. Chilled Water System: Simulation Results

5.1 System Description

The chilled water system uses the Energy Input Ratio (EIR) framework for plant operations. Efficiency is a function of chilled water supply temperature (T_{chw}) and condenser water temperature (T_{cond}). Variable-speed pumping and cooling tower fans follow affinity laws where power scales cubically with speed (P proportional to $speed^3$). The system includes: centrifugal chillers with EIR-based part-load performance curves, variable primary/secondary pumping loops, cooling tower with wet-bulb approach control, CRAH units for in-row heat rejection, and Bayesian calibration for real-world performance matching.

5.2 Annual Energy Results (8,760-Hour Simulation)

Metric	Value	Unit
Total Facility Energy	288,475	kWh
Cooling Load	40,487	kWh
Average PUE	1.163	-
Water Usage	515,331	Litres
WUE	1.786	L/kWhIT
Average COP (system)	1.000	-
Peak Cooling Load	5.853	kW
Total Carbon Emissions	158,196	kg CO2
Annual OpEx	\$42,553	USD

Table 15: Chilled Water System - Annual Energy & Emissions Summary

Note on COP: The average system COP of 1.0 reflects the conservative EIR baseline ($COP_{ref} = 3.0$ at design conditions) combined with all auxiliary loads (pumps, cooling tower fans, CRAH units). The chiller-only COP at low ambient temperatures (24 deg C) reaches 8.0, as shown in the hourly data. Real-world systems achieve system COP 4-6 at part load.

5.3 Economic Results

Metric	Value	Unit
CAPEX	\$630,000	USD
Annual OpEx	\$42,553	USD
LCCP (15-year)	\$1,838,582	USD
NPV	-\$1,092,355	USD
Simple Payback Period	50.0	Years

Table 16: Chilled Water System - Economic Analysis

The negative NPV of -\$1,092,355 and 50-year payback reflect the high CAPEX (\$630,000) of a full chilled water plant. This is typical for enterprise-scale deployments where the investment is amortised over large IT loads (MW-scale). At the simulated scale (~20-25 kW IT load), the economics are unfavourable.

5.4 Sample Hourly Performance Profile

Hour	Amb. Temp (C)	IT Load (kW)	Cool Load (kW)	Chiller (kW)	COP	Water (L)	Cost (\$)
0	24.0	21.43	2.757	2.678	8.00	43.39	2.64
1	24.0	21.43	2.757	2.678	8.00	43.39	2.64
2	24.0	21.43	2.757	2.679	7.999	43.39	2.64
3	24.0	21.43	2.758	2.679	7.998	43.39	2.64
4	24.0	21.43	2.758	2.679	7.998	43.39	2.64

Table 17: Chilled Water System - Sample Hourly Results (Hours 0-4)

5.5 Water Consumption Analysis

The chilled water system consumes 515,331 litres/year (WUE = 1.786 L/kWhIT). This is the highest water consumption of all three techniques and represents a significant operational constraint in water-scarce regions.

Region	Annual Water (L)	Scarcity Risk	Water Penalty
Seattle (low scarcity)	515,331	Low	None
Miami (moderate)	515,331	Moderate	Moderate
Denver (moderate)	515,331	Moderate	Moderate
Phoenix (high scarcity)	515,331	HIGH	Applied (2026 utility rates)

Table 18: Chilled Water System - Water Consumption by Region

5.6 Phase Gate Assessment

Gate	Status	Notes
Thermal Compliance	PASS	Inlet temps maintained within ASHRAE A2 range (18-27 deg C)
Water Constraint	PASS	515,331 L/yr within site water budget
Carbon Liability	FAIL	158,196 kg CO2/yr exceeds carbon target
Economic Viability	FAIL	NPV -\$1.09M; payback 50 years

Table 19: Chilled Water System - Phase Gate Compliance

Key Advantage: The chilled water system is the only technique to achieve full thermal compliance at ASHRAE Class A2 limits. It is the recommended choice for high-density AI workloads where server inlet temperature must be guaranteed below 27 deg C.

6. Comparative Analysis

6.1 Energy Efficiency Comparison

Metric	Air-Side Econ.	Evaporative	Chilled Water	Winner
Total Energy (kWh/yr)	478,090	141,539	288,475	Evaporative
Average PUE	1.285	1.002	1.163	Evaporative
Peak PUE	1.45	1.004	1.20	Evaporative
Cooling Overhead (%)	28.6%	0.2%	16.3%	Evaporative
Fan Energy (kWh/yr)	105,965	366	-	Evaporative
Compressor Energy	High	Zero	Moderate	Evaporative

Table 20: Energy Efficiency Metrics - All Three Techniques

6.2 Environmental Impact Comparison

Metric	Air-Side Econ.	Evaporative	Chilled Water	Winner
CO2 Emissions (kg/yr)	219,921	65,108	158,196	Evaporative
CUE (kgCO2/kWhIT)	0.591	0.461	0.549	Evaporative
Water Usage (L/yr)	0	0	515,331	Air/Evap (tie)
WUE (L/kWhIT)	0	0	1.786	Air/Evap (tie)
Carbon Tax/yr (\$)	\$27,710	\$8,270	\$20,117	Evaporative

Table 21: Environmental Metrics - All Three Techniques

6.3 Economic Comparison

Metric	Air-Side Econ.	Evaporative	Chilled Water	Winner
CAPEX (USD)	\$45,000	Low	\$630,000	Evaporative
Annual OpEx (USD)	\$97,033	\$20,523	\$42,553	Evaporative
15-yr LCCP (USD)	-	-	\$1,838,582	-
NPV (USD)	\$81,161	N/A	-\$1,092,355	Air-Side
Payback (yrs)	491.9	N/A	50.0	-

Table 22: Economic Metrics - All Three Techniques

6.4 Thermal Compliance Comparison

Metric	Air-Side Econ.	Evaporative	Chilled Water	Winner
Max Inlet Temp (deg C)	>27 (violation)	38.5 (violation)	<=27	Chilled Water
ASHRAE A1 Compliance	Marginal	FAIL	PASS	Chilled Water

Metric	Air-Side Econ.	Evaporative	Chilled Water	Winner
Airflow Violations	All 5 racks	N/A	None	Chilled Water
Throttling Events	Yes	Yes	None	Chilled Water
Cooling Failure Hours	0	0	0	Tie

Table 23: Thermal Compliance - All Three Techniques

6.5 ML Recommender System Output

The Random Forest recommender, trained on 8,760-hour simulation outputs, produces the following composite scores (higher = better overall fit for the given conditions):

Technique	Composite Score	Cost/yr	Emissions (kg)	Water (L)	Feasible
Evaporative	0.200	\$24,500	95,000	450,000	Yes
Air-Side Econ.	0.486	\$26,000	100,000	5,000	Yes
Chilled Water	0.930	\$27,235	102,084	294,088	Yes

Table 24: ML Recommender - Composite Scores

Interpretation: The ML model assigns the highest composite score to Chilled Water (0.930) because it is the only technique that satisfies all thermal constraints - the primary hard constraint for AI workloads. Evaporative cooling scores lowest (0.200) due to its thermal compliance failure, despite having the best energy and cost metrics.

6.6 Multi-Criteria Assessment Summary

Criterion	Air-Side Econ.	Evaporative	Chilled Water
Energy Efficiency	3rd	1st	2nd
Thermal Reliability	2nd	3rd	1st
Carbon Emissions	3rd	1st	2nd
Water Consumption	1st (tie)	1st (tie)	3rd
Capital Cost	2nd	1st	3rd
Operating Cost	3rd	1st	2nd
AI Workload Suitability	2nd	3rd	1st
Overall Rank	3rd	1st*	2nd

Table 25: Multi-Criteria Assessment (1st = Best)

*Evaporative ranks 1st on aggregate but fails the critical thermal compliance gate for AI workloads.

7. System Architecture

7.1 Overall Co-Simulation Architecture

The following describes the full system architecture connecting all three cooling simulation engines with the shared CloudSim Plus workload layer and the ML recommender pipeline.

```
CLOUDSIM PLUS 8.x (Workload Layer)
Hosts: Physical servers with linear power models
VMs: 1:1 mapped, TimeShared scheduler
Cloudlets: AI Training (1.8x) / Inference (1.4x) / Mixed (1.3x)
Output: hourlyITLoadKW[8760] (per-host, per-hour)
| | |
v v v
AIR-SIDE ECON EVAPORATIVE CHILLED WATER
Spring Boot Python FastAPI Spring Boot
AirEconomizerModel EvapCoolingModel ChilledWaterPhysics
Decision Tree: T_supply = EIR Framework:
FULL_ECON T_db - eta*(T_db-T_wb) COP=f(T_chw,T_cond)
PARTIAL_TRIM Lumped Capacitance: Affinity Laws:
MECH_ONLY C*dT/dt=QIT-Qcooling P ~ speed^3
| | |
v v v
RESULTS AGGREGATOR
Hourly: PUE, CUE, WUE, power breakdown, mode, violations
Annual: Energy, cost, emissions, water, phase gates
5-year: NPV, carbon tax escalation, climate scenarios
v
ML RECOMMENDER (Random Forest)
Features: energy_kwh, water_L, cost_USD, emissions_kg, violations
Output: Ranked technique + composite score + why_recommended
v
REACT FRONTEND (Vite + TypeScript)
Dashboard: Live KPIs, PUE/CUE charts, mode distribution
Simulation Detail: Hourly data tables, rack analysis
Reporting: PDF export, comparison tables
```

7.2 Chilled Water Plant Internal Architecture

```
[Cooling Tower] Wet-bulb approach control, Fan power:  $P \sim \text{speed}^3$ 
|
v
[Condenser Water Loop]  $T_{\text{cond}} = f(T_{\text{wb}}, \text{approach})$ 
|
v
[Centrifugal Chiller]  $\text{EIR} = f(T_{\text{chw}}, T_{\text{cond}})$  via bi-quadratic curves
COP_ref = 3.0 (conservative baseline)
Part-load: IPLV modelled via PLR curves
|
v
[Chilled Water Loop]  $T_{\text{chw}} = 7 \text{ deg C supply} / 12 \text{ deg C return}$ 
|
v
[CRAH Units] Variable speed fans (affinity law), Sensible heat ratio control
|
v
[Server Racks] IT Load = CloudSim hourlyITLoadKW
```

Thermal mass: $C_{\text{rack}} = 15 \text{ kJ/K}$
Inlet temp monitored vs ASHRAE 27 deg C limit

7.3 Air-Side Economizer Control State Machine

```
START (each hour)
|
v
T_out <= 24 deg C?
YES --> RH_out <= 60%?
YES --> FULL_ECON: OA=100%, P_comp=0
P_fan = V_req x W/CFM x 1.15 (filter penalty)
NO --> PARTIAL_TRIM: OA=20%, P_comp = Q_mech/COP
NO --> MECHANICAL_ONLY: P_comp = Q_IT/COP(T_out)
|
v
Airflow Violation Check: V_req > V_max?
YES --> VIOLATION: Recommend liquid cooling (direct-to-chip)
|
v
Compute: PUE, CUE, hourly cost, carbon emissions
```

8. Climate Resilience and Sustainability Analysis

8.1 IPCC Climate Scenario Impact

Performance is modelled against three IPCC RCP pathways for 2030. A 3% cooling load increase per 1 deg C temperature rise is applied. Under RCP 8.5, the Air-Side Economizer is most severely impacted because higher ambient temperatures reduce free-cooling hours, forcing more mechanical operation.

Scenario	Temp Offset	Air-Side (kWh)	Evap. (kWh)	Chilled W. (kWh)
RCP 2.6 (Baseline)	+0 deg C	478,090	141,539	288,475
RCP 4.5 (Moderate)	+1.5 deg C	499,604	147,909	301,456
RCP 8.5 (High)	+3.0 deg C	521,118	154,278	314,438

Table 26: Climate Scenario Impact on Annual Cooling Energy

8.2 Carbon Tax Escalation (15% Annual)

Year	Air-Side Econ.	Evaporative	Chilled Water
Year 1	\$27,710	\$8,270	\$20,117
Year 2	\$31,867	\$9,511	\$23,135
Year 3	\$36,647	\$10,937	\$26,605
Year 4	\$42,144	\$12,578	\$30,596
Year 5	\$48,466	\$14,465	\$35,185
5-yr Total	\$186,834	\$55,761	\$135,638

Table 27: Carbon Tax Cost Projection (15% Annual Escalation, IPCC 2030)

8.3 Water Stress Assessment

Technique	Annual Water (L)	WUE	Phoenix Risk
Air-Side Economizer	0	0.0	None
Evaporative (IEC)	0	0.0	None
Chilled Water	515,331	1.786	HIGH - Water Penalty Applied

Table 28: Water Stress Assessment by Technique and Region

9. Discussion and Conclusions

9.1 Key Findings

1. Evaporative Cooling achieves the best energy efficiency (PUE 1.002) and lowest operating cost (\$20,523/yr) but fails thermal compliance for AI-density workloads. Rack inlet temperatures of 38.5 deg C exceed the ASHRAE Class A1 limit by 11.5 deg C, triggering aggressive CPU throttling.
2. Chilled Water is the only thermally compliant solution for high-density AI workloads. It maintains inlet temperatures within ASHRAE A2 limits (18-27 deg C) at all hours. However, its high CAPEX (\$630,000) and water consumption (515,331 L/yr) are significant constraints.
3. Air-Side Economization is unsuitable as a standalone solution for AI workloads. All five racks exhibit airflow violations (required CFM 2-4x the per-rack limit), and the system's high carbon emissions (219,921 kg CO2/yr) result in a prohibitive carbon tax burden (\$27,710/yr).
4. The ML recommender correctly identifies Chilled Water as the highest-scoring technique (0.930) when thermal compliance is weighted as a hard constraint, despite Evaporative having better energy and cost metrics.
5. Carbon tax escalation disproportionately penalises Air-Side Economization. Over 5 years, its cumulative carbon tax (\$186,834) is 3.4x higher than Evaporative (\$55,761).

9.2 Recommended Hybrid Strategy

Based on the simulation results, a hybrid Evaporative + Chilled Water configuration is recommended:

- Deploy evaporative pre-cooling to reduce condenser water temperature entering the chiller.
- Use chilled water as the primary cooling loop to guarantee ASHRAE compliance.
- This hybrid approach can achieve PUE ~1.05-1.10 while maintaining full thermal compliance.
- Estimated water savings: 30-40% vs. standalone chilled water.

9.3 Limitations

- The chilled water COP of 1.0 is conservative; real-world systems achieve COP 4-6 at part load.
- The evaporative simulation uses a constant ambient temperature (25 deg C) rather than TMY3 data.
- Air-side economizer results use TMY3 weather data but a simplified constant COP = 3.0 for mechanical backup.
- All simulations assume a single-zone data centre; multi-zone hot/cold aisle containment would improve all results.

9.4 Future Work

- Implement enthalpy-based economiser control (replacing dry-bulb/RH decision tree).

- Integrate real EIR part-load curves from manufacturer datasheets for chiller modelling.
- Extend the ML recommender to include hybrid configurations as candidate solutions.
- Validate simulation results against real data centre measurements (ASHRAE TC 9.9 benchmarks).
- Model multi-site deployment across Phoenix, Miami, Denver, and Seattle using full TMY3 datasets.

Appendix A: Governing Equations Reference

Equation	Formula	Used In
Heat Balance	$C_{total} \cdot dT/dt = Q_{IT} - Q_{cooling}$	All three
Airflow Requirement	$V_{req} = (P_{IT_kW} \times 3160) / (\rho \times C_p \times \Delta T)$	Air-Side, Evap
DEC Supply Temp	$T_{supply} = T_{db} - \eta \times (T_{db} - T_{wb})$	Evaporative
Fan Power	$P_{fan} = V_{CFM} \times W/CFM \times f_{filter}$	Air-Side, Evap
Mechanical Cooling	$P_{mech} = Q_{mech} / COP$	Air-Side, Chilled W
Affinity Law	P proportional to speed ³	Chilled Water
PUE	$PUE = P_{total} / P_{IT}$	All three
CUE	$CUE = (P_{total} \times CI) / P_{IT}$	All three
WUE	$WUE = \dot{m}_{water} / P_{IT}$	Chilled Water
Throttling	$\Phi_{adj} = \Phi_{base} \times (1 - \gamma \times \Delta T)$	All three
NPV	$NPV = \sum (R_t / (1+i)^t) \text{ for } t=1..15$	All three
Workload Model	$U(t) = [U_{diurnal}(t) \times M_{AI}] + \sigma(t)$	All three
UPS Efficiency	$\eta = -0.05u^2 + 0.1u + 0.94$	All three

Table A1: Complete Governing Equations Used Across All Three Simulations

Constants: $\rho = 1.2 \text{ kg/m}^3$, $C_p = 1.006 \text{ kJ/(kg.K)}$, $L_v = 2,260 \text{ kJ/kg}$, $i = 0.0971$ (discount rate), $\gamma = 0.05/\text{deg C}$ (27-29 deg C range), $\gamma = 0.10/\text{deg C}$ (29-32 deg C range).